Google Play Store

Exploratory Data Analysis (EDA)

Data Source: Kaggle (Google Play Store Apps)

Dataset Size: 10,841 rows and 13 columns

Data Type: Structured tabular dataset (CSV format)

Tools Used: Python (Pandas, Seaborn, Matplotlib)

1. Introduction:

This project performs an exploratory analysis of Google Play Store app data. The workflow includes initial inspection, basic cleaning, and descriptive visualizations to highlight market patterns. The dataset contains 10,841 rows and 13 columns, including several attributes such as app name, category, type, and price.

2. Data Preparation and Cleaning:

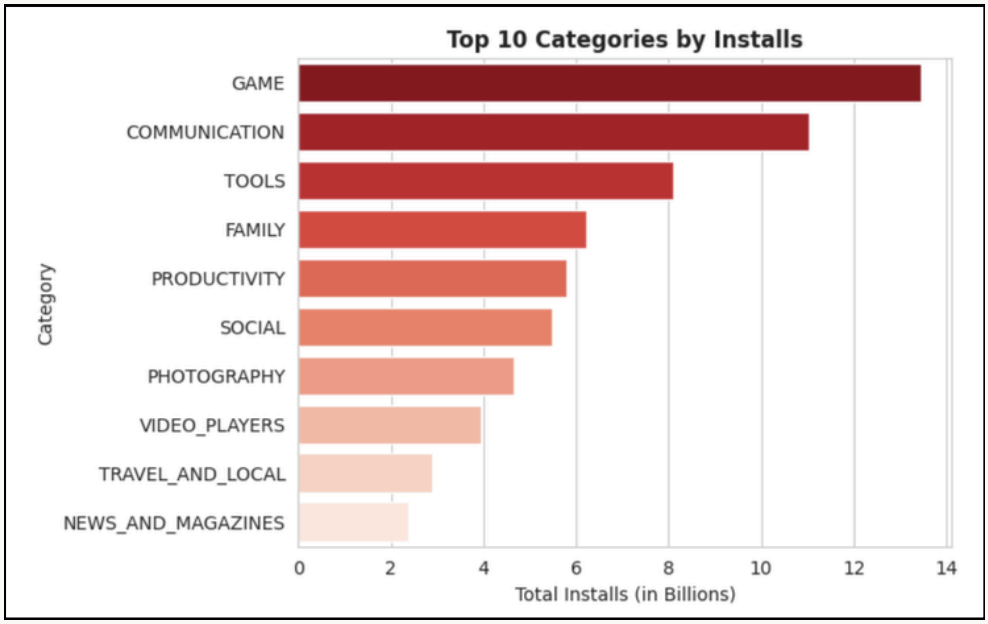
- Importing libraries (Pandas, Seaborn, Matplotlib) and loading the data into a DataFrame.
- Exploring the data using exploration methods such as *head(), shape, info(),* and *describe().*
- Converting column names to lowercase and replacing spaces with underscores.
- Handling duplicated data by converting the *reviews* column to numeric and sorting the data by it, then dropping duplicated apps while keeping the first occurrence.
- Removing data entry errors and handling missing values.
- Cleaning special characters and transforming columns into appropriate formats.
- **Final dataset size**: The data was refined from 10,841 records to 8,196 clean rows.

Full implementation details and source code are available on [GitHub](#).

3. **Data Analysis & Visualization:**

3.1 **Category Popularity based on Installs (Top 10):**

Question: Which categories attract the highest number of installs?



Note: Install values were originally provided as minimum install thresholds (e.g., 1,000,000+). The '+' symbol was removed during cleaning, so results represent approximate minimum counts

The chart shows that the Game category leads in total installs, followed by Communication and Tools. This indicates strong user demand for gaming and communication applications within the Google Play Store.

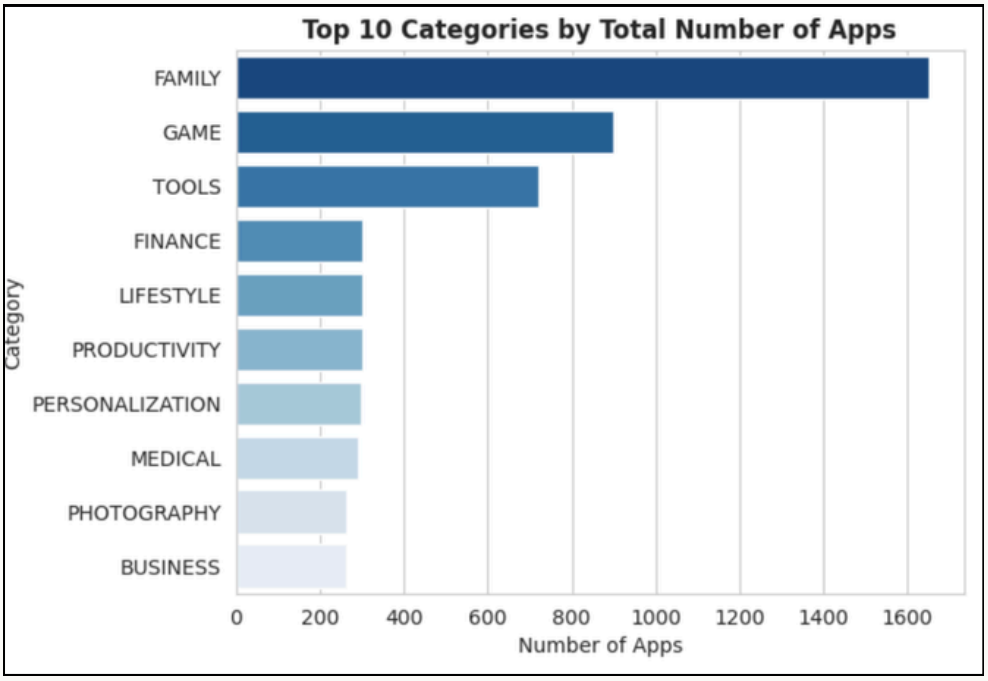
```
category_installs= df.groupby('category')['installs'].sum().sort_values(ascending=False).reset_index().head(10)
x_data = category_installs['installs'] / 1e9
sns.set_style('whitegrid')
a=sns.barplot(data=category_installs, x= x_data , y='category', hue='category', palette='Reds_r')
a.set_title('Top 10 Categories by Installs',fontweight='bold')
a.set(xlabel= ('Total Installs (in Billions)'), ylabel= 'Category')

plt.show()
```

The data was grouped by category, aggregated using the sum of installs, and sorted to extract the top 10 categories.

3.2 **Number of Apps per Category (Top 10):**

Question: How is the app market distributed across categories?



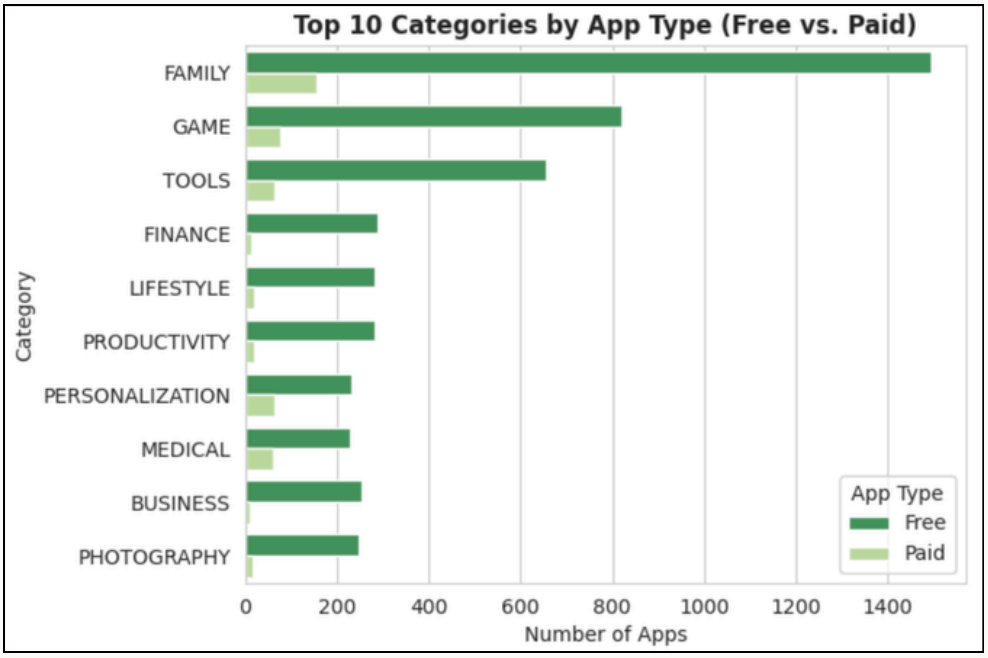
Unlike the previous chart, this chart shows that the Family category contains the highest number of applications. This indicates that high supply does not necessarily mean high demand. The Game category, despite having fewer apps than Family, achieves significantly higher total installs, suggesting stronger user engagement and demand concentration.

```
category_apps= df.groupby('category')['app'].count().sort_values(ascending=False).reset_index().head(10)
b=a=sns.barplot(data=category_apps, x= 'app' , y='category', palette='Blues_r')
b.set_title('Top 10 Categories by Total Number of Apps',fontweight='bold')
b.set(xlabel= ('Number of Apps'), ylabel= 'Category')
plt.show()
```

The data was grouped by category and aggregated using the count of applications to determine the top 10 categories by number of apps.

3.3 **App Type Distribution by Category (Top 10):**

Question: How are free and paid apps distributed within each category?



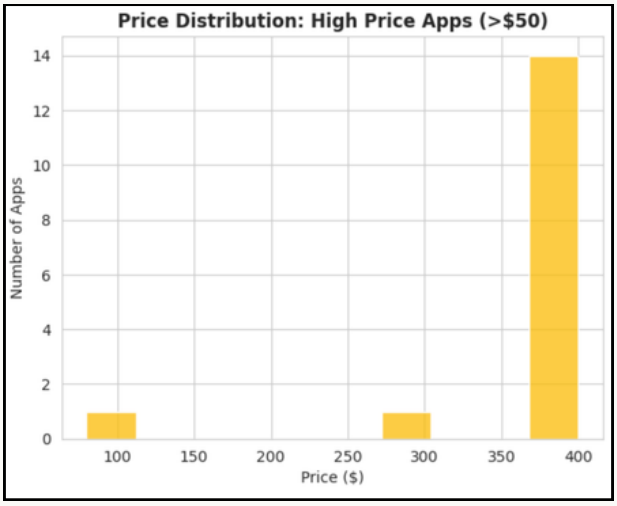
The chart shows that free applications dominate across all categories. Paid apps represent only a small proportion within each category, even in high-volume segments such as Family and Game. This suggests that most apps are free, and developers earn revenue through ads or in-app purchases rather than charging users before installation.

```
top_10_categories = df['category'].value_counts().head(10).index
top10=df[df['category'].isin(top_10_categories)]
c=sns.countplot(data=top10, y='category', hue='type', palette='YlGn_r', order=top_10_categories)
c.legend(title='App Type')
c.set_title('Top 10 Categories by App Type (Free vs. Paid)',fontweight='bold')
c.set(xlabel= 'Number of Apps', ylabel= 'Category')
plt.show()
```

The top 10 categories were identified based on frequency using value_counts(). The dataset was then filtered to include only these categories, and a countplot was used to compare the distribution of free and paid apps within each category.

3.4 **Low vs. High Price Distribution:**

Question: What is the distribution of prices among paid apps?



Here, the data was separated into low- and high-price ranges to improve visualization clarity, as outlier values compressed the overall distribution when plotted together. The two charts indicate that most paid apps are priced between $1 and $5, followed by a smaller cluster between $6 and $10. High-priced apps are rare, suggesting a preference for affordable pricing strategies.

```
low_price_apps = df[(df['type'] == 'Paid') & (df['price'] <= 50)]
d=sns.histplot(low_price_apps['price'], bins=20, color='#34A853')
d.set_title('Price Distribution: Low Price Apps (<=$50)', fontweight='bold')
d.set(xlabel= 'Price ($)', ylabel= 'Number of Apps')
plt.show()
```

```
high_price_apps = df[(df['type'] == 'Paid') & (df['price'] > 50)]
e=sns.histplot(high_price_apps['price'], bins=10, color='#FBBC05')
e.set_title('Price Distribution: High Price Apps (>$50)', fontweight='bold')
e.set(xlabel= 'Price ($)', ylabel= 'Number of Apps')
plt.show()
```

Paid apps were filtered and split into low- and high-price groups using a $50 threshold, and histograms were used to visualize their price distributions.

4. **Conclusion:**

Overall, the analysis highlights a clear imbalance between supply and demand across categories. While the Family category contains the largest number of apps, the Game category dominates in total installs. In addition, the market is largely free-based, and even paid apps tend to follow low-price strategies. These patterns suggest that the Google Play Store market is largely accessibility-driven, with free access and low pricing being common across categories.

This exploratory analysis provides initial insights into category popularity, app type distribution, and pricing patterns. Further investigation could examine installs per app, revenue models, and temporal trends for deeper understanding.